GEMS INTERNATIONAL SCHOOL — DUBAI HILLS

Centre of Excellence

Summer Camp 2026 — Programme

Dates: 13th – 17th July 2026
Venue : Centre of Excellence, GEMS International School — Dubai Hills

Program Structure

The five-day program bridges conceptual learning with practical execution through two distinct phases:

- **Days 1–4: Hands-On STEM Rotations**
  Students rotate through a series of practical, project-based modules, including **CubeSat**, **Magician Go Robotics**, **Drone Flying**, **Obstacle Avoidance**, **Straw Rocket Design**, and **Flight Simulation**.
- **Day 5: Exhibition & Presentation Showcase**
  The final day is dedicated to a live showcase. Students display their work, demonstrate their core learning outcomes, and present their finished projects to parents, staff, and invited guests.

Core Objectives

The camp directly drives the Centre of Excellence vision by cultivating vital future-ready skills:

- **Innovation & Creativity:** Applying emerging technologies to solve modern problems.
- **Collaboration:** Working in teams to build and test engineering designs.
- **Real-World Application:** Transforming STEM theories into tangible, functional projects.

Programme Overview

Parameter	Details
Duration	5 Days
Daily Timings	9:00 AM – 3:00 PM
Break	45-minute break during the day
Daily Structure	Two main learning sessions per day
Target Students	Upper Primary and Secondary students
Delivery Model	School-led practical STEM camp
Final Outcome	Exhibition, project display, and student presentation

Project Areas

1. CubeSat Project

Students will explore the concept of miniature satellites and learn how CubeSats are used in real-world space missions. They will investigate basic satellite structures, mission objectives, payload ideas, and design considerations.

Students will experience:

- Introduction to CubeSat and space technology
- Basic satellite mission planning
- Design of a CubeSat model
- Understanding payload, sensors, and communication concepts
- Team-based problem-solving and presentation preparation

2. Magician Go Robotics

Students will work with the Dobot Magician Go platform to understand robotics movement, automation, navigation, and task-based programming. This project will allow students to explore how robots can be used in logistics, transportation, and smart automation systems.

Students will experience:

- Introduction to mobile robotics

- Robot movement and control
- Path planning and navigation challenges
- Object transportation simulation
- Automation-based problem-solving

3. Drone Flying

Students will learn the fundamentals of drone technology, safe flying practices, and the role of drones in modern industries such as logistics, security, media, agriculture, and disaster response.

Students will experience:

- Introduction to drone components and controls
- Basic drone flying skills
- Safe operation and supervision protocols
- Understanding balance, thrust, lift, and direction
- Practical flying challenges

3a. Obstacle Avoidance Challenge

As part of the drone and robotics experience, students will complete an obstacle avoidance challenge. They will learn how machines detect obstacles and respond to their environment.

Students will experience:

- Understanding sensors and obstacle detection
- Designing simple obstacle courses
- Testing drone or robot movement through challenges

3b. Straw Rocket Design

Students will participate in an aerospace engineering activity where they design, build, test, and improve straw rockets. This will help them understand aerodynamics, force, motion, stability, and design iteration.

Students will experience:

- Rocket design and construction
- Testing launch distance and accuracy
- Understanding fins, nose cones, weight, and stability

- Recording results and improving designs
- Applying engineering design cycle principles

3c. Flight Simulation

Students will use flight simulation activities to understand the basics of aircraft control, navigation, take-off, landing, and flight decision-making.

Students will experience:

- Introduction to flight controls
- Understanding pitch, roll, yaw, and throttle
- Simulated take-off and landing practice
- Basic navigation and flight awareness
- Reflection on real-world aviation and technology careers

Daily Programme Structure

Day 1 – CubeSat and Drone Flying

Students will begin the camp with an introduction to the CubeSat project and drone flying. They will explore the basics of satellite design, mission planning, drone safety, and basic flight controls.

Focus Areas:

- Camp orientation and safety briefing
- Introduction to CubeSat and space technology
- CubeSat mission planning and basic design
- Introduction to drone components and controls
- Basic drone flying practice

Day 2 – CubeSat and Magician Go

Students will continue developing their CubeSat project and begin working with the Magician Go robotics platform. They will explore robotics movement, automation, and task-based challenges.

Focus Areas:

- CubeSat model development
- Payload and design improvement

- Introduction to Magician Go robotics
- Robot movement and navigation
- Object transportation simulation

Day 3 – Drone Flying and Magician Go

Students will develop their drone flying skills and continue working with Magician Go. They will complete obstacle-based challenges and apply problem-solving skills through robotics and flight activities.

Focus Areas:

- Drone flying skill development
- Obstacle avoidance challenge
- Magician Go programming and control
- Robotics path planning
- Testing, troubleshooting, and improvement

Day 4 – CubeSat and Drone Flying

Students will finalise their CubeSat models and complete advanced drone flying activities. They will also begin preparing their project displays and presentations for the final exhibition.

Focus Areas:

- CubeSat final build and refinement
- Drone flying challenge completion
- Flight simulation and aerospace connection
- Project display preparation
- Presentation planning and rehearsal

Day 5 – Exhibition and Student Presentation

The final day will be dedicated to celebrating student learning. Students will display their completed work, demonstrate their projects, and present their learning journey to parents, staff, and invited guests.

Focus Areas:

- CubeSat project display
- Magician Go robotics demonstration

- Drone flying and obstacle challenge showcase
- Straw rocket and flight simulation display, where applicable
- Student team presentations
- Certificates and recognition ceremony

Learning Outcomes

By the end of the camp, students will be able to:

- Understand basic concepts in robotics, drones, aerospace, and automation
- Apply engineering design thinking to practical challenges
- Build, test, and improve working models
- Demonstrate safe and responsible use of technology
- Develop teamwork, creativity, communication, and problem-solving skills
- Present their ideas confidently during the final exhibition
- Connect classroom learning with real-world STEM applications

Exhibition and Celebration

The final exhibition will include:

- CubeSat model display
- Magician Go robotics demonstration
- Drone flying and obstacle challenge showcase
- Certificates of participation